C. K-Dominant Character

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: binary search, implementation, two pointers | https://codeforces.com/problemset/problem/888/C

You are given a string *s* consisting of lowercase Latin letters. Character *c* is called *k*-dominant iff each substring of *s* with length at least *k* contains this character *c*.

You have to find minimum *k* such that there exists at least one *k*-dominant character.

Input

The first line contains string *s* consisting of lowercase Latin letters ($1 \le |s| \le 100000$).

Output

Print one number — the minimum value of *k* such that there exists at least one *k*-dominant character.

Examples

Input
abacaba

Output
2

Input
zzzzz

Output
1

Input
abcde

Output
3